Name	Eng POCs	Metrics Impact (SSv2, HF, Fresh, MAU/DAU), if any	Cost Savings, if any	Foundational Impact (3 sentences max in bullet form), if any	LR Doc	Link to Slack Proposal and Discussion	(WIP) Notes
Conditional Learned Retrieval (CLR) on Recommended Interests + Unity Migration	Bowen Deng Hongtao Lin	Total Successful Session(SSv2): +0.22% P6: +0.46% Top funnel successful sessions +0.21% P6: +0.30% Mid funnel successful sessions +0.24 % P6: +0.39%	This launch also deprecates major hf usecases in interest service and manas, and it can save >$350k for interest service deprecation		[LR] hf_conditional_retrieval_recommended_interests + unity migration	https://pinterest.slack.com/archives/C05UMECTDDJ/p1736186706302699	
OmniSage + UVE features for Learned Retrieval	Yuke Yan Bowen Deng	Intent Expression Only Sessions: +0.12% Click-out Sessions: +0.85% HF Closeups: +0.41% HF Clicks: +0.67% HF LongClicks: +1.69%			[LR] Homefeed L2R: Omnisage + Unified Visual Embedding V4	https://pinterest.slack.com/archives/C05UMECTDDJ/p1736290438322359	
Ray Adoption and Mixed Precision for LR and CLR	Bowen Deng zhibofan@pinterest.com	N/A	$30k	• Modernize training stack, enable ray-trainer joins and fast ML infra • 80% training throughput improvements with ray and mixed precision	[LR] Homefeed Learned Retrieval + Conditional Leaned retrieval Ray Adoption and Mixed Precision	https://pinterest.slack.com/archives/C05UMECTDDJ/p1736814345772019	
Conditional Learned Retrieval (CLR) for BMI	Devin Kreuzer Hongtao Lin Bowen Deng	Successful Sessions: Total Successful Sessions (SSv2): +0.11% Save, Revisitation, Social and Download Sessions (SRSD SSv2): +0.13% Top Funnel Successful Sessions (SSv1): +0.11% Mid Funnel Successful Sessions (SSv1): +0.18% Shopping Save, Revisitation, Social and Download Sessions (SSv2): +0.21% Shopping Closeup Sessions, Non-MECE (SSv2): +0.25% Total 6-box Successful Sessions (SSv1): +0.12% Total 6-box Unsuccessful Sessions (SSv1): +0.17% Board More Ideas: Time Spent: +2.11% Sessions: +0.70% Impressions: +0.69% Repins: +3.79% Closeups: +4.65%	$120k cost increase	First successful application of CLR outside Interest CLR on grid. Established the entire objective of expanding CLR use cases and iterating on modelling for the rest of 2025.	[LR]: Board Conditional Learned Retrieval on BMI	https://pinterest.slack.com/archives/C05UMECTDDJ/p1736873003693229	
Manas Quantization for LR and CLR	Bowen Deng emadsen@pinterest.com Bowen Zhou	N/A	$225k	unblock us from further scaling our retrieval system to a much larger serving corpus and larger embeddings	[LR]HF Learned Retrieval and Conditional Learned Retrieval Manas Quantization	https://pinterest.slack.com/archives/C05UMECTDDJ/p1737490967725759	
Multi-Embedding to replace Single-Embedding Learned Retrieval	Yuke Yanzhibofan@pinterest.com Bowen Deng	All Users: Intent Expression Only Sessions: +0.30% Core Users: Repins + 0.51% Intent Expression Only Sessions +0.35% Non-shopping Click-out sessions: +0.65%		New paradigm for two-tower retrieval that captures distinct user interests, unlocking new opportunities for modeling.	LR: [HF] Multi-Embedding Learned Retrieval via Differentiable Clustering	https://pinterest.slack.com/archives/C05UMECTDDJ/p1737572814741049	
Chunk Invalidation Actions Deprecation	J.J Hu Dafang He	N/A	$180k cost saving	• Simply chunk invalidation actions only through repin, hide, report	LR: homefeed chunk invalidation revisit	https://pinterest.slack.com/archives/C05UMECTDDJ/p1738019384245619	
Auto Retraining Framework for Lightweight Scoring	Cheng Duan Bella Huang	0.11% Top Funnel (Total Successful Sessions (SSv2)) +0.37% Homefeed Timespent +0.25% Homefeed Impressions, +0.99% Homefeed Repins, +0.60% Homefeed Closeups		• Keep the model refresh with the latest training data	[HF] LWS Auto Retrain (ARF)	https://pinterest.slack.com/archives/C05UMECTDDJ/p1736383505853879	
Ray Adoption for LWS	Piyush Maheshwari	Shopping Click Sessions (SSv2) +0.30% HF Closeup +0.30%	$10k	• Modernize training stack, enable ray-trainer joins and fast ML infra • Training throughput: 138k examples/sec -> 305k examples per sec • Training time: 17 hours -> 8.5 hours	LR: HF LWS Ray Adoption	https://pinterest.slack.com/archives/C05UMECTDDJ/p1738348976419399	
SOAP optimizer for LWS	Abhinav Naikawadi Piyush Maheshwari	SSV2 +0.19% Intent Expression Only Sessions: +0.28% Pin Impressions: +0.27% Closeup: +0.46% 5s+ Outbound Clicks: +0.48% Sessions > 5min: +0.43% Pins 28d Fresh Impressions: +0.26% Trustworthy Pins 28d Fresh Impressions: +0.45% Homefeed Pin Impressions: +0.22% Homefeed Num Pin Click: +1.39%			[LR]: Homefeed LWS SOAP optimizer	https://pinterest.slack.com/archives/C05UMECTDDJ/p1741970550753649	
L1 Calibration	Matthew Lawhon Piyush Maheshwari Aditya Mantha	N/A	N/A	• Significantly higher model head stability (measure by head importance) • Improved model score interpretability • Improved L1/L2 Alignment	LR: L1 Calibration	https://pinterest.slack.com/archives/C05UMECTDDJ/p1742241633880649	
Deprecate Long Impression Distillation in LWS	Piyush Maheshwari	- Total Successful Sessions (ssv2) 0.12% - Repins Overall 0.18% (Unique Users) - Pins 28d Fresh Repins 0.74% - Pins 28d Fresh Neardup Impressions without Ads 0.25% - PIns 28d Fresh Neardup Repins without Ads 1.04% - Homefeed Reports -14.67% Overall, -22.95% P6 - Related Repins Diluted 0.25% (unique users)	N/A	• Improve L1/L2 Alignment	[LR] HF LWS Deprecate Long Impression Distillation	https://pinterest.slack.com/archives/C05UMECTDDJ/p1743706304255179	
Removes Long Impression Training Objective in LWS	Piyush Maheshwari	Homefeed pin report -10.92 % Downloads on HF 0.59%	N/A	• Improve L1/L2 Alignment	[LR] HF LWS Deprecate Long Impression Training Objective	https://pinterest.slack.com/archives/C05UMECTDDJ/p1749240681217589	
OmniSage + UVE features for Lightweight Scoring	Devin Kreuzer Piyush Maheshwari	SSV2 metrics: +0.11% Total Successful Sessions SSV2 +0.18% SRSD SSV2 +0.63% Overall Repins +0.32% Overall 5s+ Outbound Clicks +0.58% Overall Board Creates +0.18% HF Impressions +0.26% HF Closeups +0.73% HF Repins +0.93% HF Longclicks	N/A	• Improve L1/L2 Alignment	[LR] LWS Omnisage	https://pinterest.slack.com/archives/C05UMECTDDJ/p1741654477038219	
Apiary Index Reduction	Sona Shah J.J Hu Lucy Qiao	N/A	$360k cost saving	• Reduce Apiary index size by filtering out boards with repins in the last 28 days	LR: hf_apiary_index_reduction_exp	https://pinterest.slack.com/archives/C05UMECTDDJ/p1742410424419839	
L1 Serving Optimization	Devin Kreuzer J.J Hu Matthew Lawhon Piyush Maheshwari	-0.09% HF Pin Impressions +0.42% HF Clickthroughs +0.57% HF Longclickthroughs +0.7% HF PWT	$345k cost saving	• Use mini-batch (with fixed batch size) instead of request per feedsource for pinnability inference request	LR: L1 Serving Optimization	https://pinterest.slack.com/archives/C05UMECTDDJ/p1741107892271839	
LWS Online Alignment Metrics	Hedi Xia Piyush Maheshwari Bella Huang	N/A	N/A	• Proved both theoretically and experimentally how evaluation should be done for pre-ranker (LWS) models • Provide better understanding of LWS model tradeoffs during online experiments • The evaluation framework is useful to all pre-ranker models in core/disco teams	[LR] Light Weight Scoring Online Alignment Metrics	https://pinterest.slack.com/archives/C05UMECTDDJ/p1747244009848059	
Offline Replay for Accurate LWS Offline Evaluation	Hedi Xia Dylan Zhou Tianyou Wang Henry Feng	N/A	N/A	• Bring the accurate LWS online alignment metrics offline to a scalable and easy-to-use solution • Provide reliable forecast of metric shifts quantitatively for the first time in LWS • The evaluation framework is useful to all retrieval stage models in core/disco teams, a wider range than just pre-ranker models	[LR] Offline Replay for Accurate LWS Offline Evaluation	https://pinterest.slack.com/archives/C05UMECTDDJ/p1762371808654119	
Pairwise Loss in LWS	Hedi Xia Yichu Zhou (He Him) Hongtao Lin Piyush Maheshwari	Total Successful Sessions: +0.09% Intent Expressions Only Sessions, MECE: +0.25% 28d Fresh Impressions: +0.43% 28d Fresh Repins: +0.28% 28d Fresh Impressions without ads: +0.71% (Unique Users) Web or API request +0.01% (Unique Users) Days active in last 7 days (WAU) +0.01%	N/A	• Improve L1/L2 Alignment • Enable easier adoptions of sparse heads in the future	[LR] Homefeed Light Weight Scoring Pairwise Loss	https://pinterest.slack.com/archives/C05UMECTDDJ/p1744914999260489	
LWS GPU Serving	Piyush Maheshwari Matthew Lawhon	Total Success Sessions : +.17% SRSD Sessions .22% Closeup .26% HF Pin Click .86% HF downloads .91%	N/A	• Enables step function scale up up, next gen model architecture iteration of early stage ranker	[LR] HF LWS GPU Serving and Model Scaleup	https://pinterest.slack.com/archives/C05UMECTDDJ/p1761252306724919	
Model Farm Migration for Learned Retrieval	Yuke Yan Bowen Deng	Neutral metrics. 4X Model Capacity for Large ID Embedding Model 10 --> 44		4X Model Capacity = 4X experiment velocity. Model iterations with large ID embedding only become possible at this point, previously we have to remove ID embeddings to save space.	[LR] Homefeed Learned Retrieval Model Farm Migration	https://pinterest.slack.com/archives/C05UMECTDDJ/p1738870441029699	
Trustworthy product signal migration from PIS to IPG	David WangJ.J Hu	BMI: non-sig HFNP: Intent Expressions Only Sessions Users +0.09% Pin Searches Users +0.08%	5.2k cost increase	• Use product trustworthy signal from IPG instead of PIS for HFNP and BMI	[LR] hfnp_ipg_migration [LR] bmi_ipg_migration	https://pinterest.slack.com/archives/C05UMECTDDJ/p1739223060431809	
BMI Graph Product Pin CG deprecation	Devin Kreuzer smalreddy@pinterest.com	N/A	50k Cost savings	None, permits easier BMI migration to GULP	[LR] Homefeed BMI Graph Product Pin Deprecation	https://pinterest.slack.com/archives/C05UMECTDDJ/p1740158526892569	
Shopping Corpus Iterations for Diverse Merchants	Wei-Ting Lin J.J Hu Mostafa Keikha Chao Wang	Total Successful sessions (SSv2) +0.12% Intent Expressions Only sessions, MECE (SSv2) +0.19% +0.91% TMPL impressions TML excl Etsy imps share: +1.46% Golden Path Shoppers (GPS) (% of WAU): +0.16% to +0.46% SID on Top 5 TWPP: +0.4%	N/A	• Add more items from tail/torso merchants to the two (TML and Regular) shopping corpora that improves merchant diversity	[LR] hf_shopping_corpus_mcd_v2	https://pinterest.slack.com/archives/C05UMECTDDJ/p1741884764749959	
CG Tuning for Revamp NUX	Zihao Chen Lingzhi Luo	Sessions: SSV2 sessions +0.91% Intent-only sessions +1.06% SRSD: +0.78% Shopping clickout sessions: +3.06% HF Actions:	N/A	N/A	[LR] CG budget tuning on NUX Revamp non-day0 users	https://pinterest.slack.com/archives/C05UMECTDDJ/p1741986856092679	
Tuning Conditional Dropout in CLR	Devin Kreuzer	BMI Timespent (Propensity): +0.75% (-0.19%) BMI Closeups (Propensity): +0.60% (+0.25%) BMI Repins (Propensity): +0.94% (+0.32%) US Market: +2.23% BMI Outbound Clicks (US Market): + 1.43% Homefeed Tabs Landing Page Time Spent: +0.83% Homefeed Tabs Landing Page Repins: +1.36% US Market: +2.85%	N/A		LR: Tuning Condition Dropout in CLR	https://pinterest.slack.com/archives/C05UMECTDDJ/p1742854577390499	
Coterie Features in Learned Retrieval	Zihao Chen Yuke Yan Bella Huang	Sessions: Total Successful Sessions: +0.03% Non Shopping Click Out Sessions, MECE (SSv2): +0.69% HF Actions	N/A	N/A	[LR] User & Pin Coterie features in Homefeed Learned Retrieval Model	https://pinterest.slack.com/archives/C05UMECTDDJ/p1744217037984669	
Offsite Expansions to Shopping LR Corpus	J.J Hu Wei-Ting Lin Chao Wang	Intent Expressions Only Sessions, MECE(SSv2) +0.13% Shopping Click out Sessions (SSv2) +0.2% WAU global +0.08%, U.S +0.17% Repin propensity in UCAN +0.18% TMPL (Excl. Etsy) Impressions (Homefeed, Non-ad): +3.47% TMPL (Excl. Etsy) 5s+ Outbound Clicks (Homefeed, Non-Ad): +1.45%	$2.3k cost increase	• Add new conversion item (with checkout event) into shoppng LR corpus	[LR] hf_shopping_offsite_corpus	https://pinterest.slack.com/archives/C05UMECTDDJ/p1744925845556009	
Item Pin Selection Changes for Shopping LR Corpus	Wei-Ting Lin J.J Hu Sai Kiran Nelluri	TP Fresh Impressions: +0.4% / 0.09% TP Fresh repins: +0.31% / +0.96% Trustworthy Pins 28d Fresh Neardup Impressions without Ads: +0.67% / +0.21% Pin Report: -7.85% / -0.95% (Homefeed) Trustworthy Pins 28d Fresh Impressions without Ads: +5.79% / +1.23% (Homefeed) Trustworthy Pins 28d Fresh Closeups without Ads: +2.76% / +2.16% (Homefeed) Trustworthy Pins 28d Fresh Repins without Ads: +3.23% / +2.21%	N/A	• Switch to the IPS V7 (the current production item-pin-selection version), which prioritize "life-style" images within an item	[LR] hf_shopping_corpus_ips_v7	https://pinterest.slack.com/archives/C05UMECTDDJ/p1744994460115069	
L1 Serving Optimization for BMI and HFNP	Zili Li Devin Kreuzer	N/A	N/A	N/A	LR: L1 Serving Optimization: All Unity-HF Surfaces	https://pinterest.slack.com/archives/C05UMECTDDJ/p1745594528225029	
Multi-Embedding ARF + ID Embeddings	Yuke Yan Bowen Deng	Save, Revisitation, Social and Download Sessions (SSV2): +0.14% Site-wide Repins: +0.32% Board Creates: +0.61% HF Repins: +0.87% HF Long Clickthroughs: +0.82% Use Case Adoption: +0.26% Use Case Retention: +0.34% Homefeed WAU with 2+ use cases: +0.15%	$236k cost saving	Fully launched Multi-Embedding in Prod and realize the cost savings. Revamp ARF such that we retrain every 3 days, ensuring the model always captures the user's latest behaviors (as evident by the board create, repin increases).	[LR] Multi-embedding Retrieval: ARF + Latest Features	https://pinterest.slack.com/archives/C05UMECTDDJ/p1746574717967769	
Unified Interest ID + Board ID CLR	Devin Kreuzer Hongtao Lin Bowen Deng	Total Successful Sessions (SSv2): +0.09% SRSD v2: +0.12% Total Time Spent on App: +0.16% Users > 15 min: +0.23% Users > 30 min: +0.30% Homefeed WAU with 2+ use cases: +0.22% Pin Impressions: +0.13% Downloads: +0.38% Closeups: +0.21% Sessions > 5 min : +0.19% HF Longclicks: +1.16% BMI Closeups: +1.00% BMI Board Section Creates: +2.99%		Single Unified CLR model which can onboard more conditions and be applied to many more use cases across Pinterest surfaces 70% increase training speed with Conditional Filtering applied (more on this in the doc) At least 858 dev hours / 21 full-time working eng weeks saved from maintaining one model instead of 4+ (2 existing, at least 2+ planned for near future)	[LR]: Unified CLR v0	https://pinterest.slack.com/archives/C05UMECTDDJ/p1747156807722349	
Offsite PinnerSage v2 Pixie CG	Lingzhi Luo Dafang He				LR: offsite pinnersage v2 pixie candidate generator LR: offsite pinnersage v2 pixie candidate generator for EU user	https://pinterest.slack.com/archives/C05UMECTDDJ/p1747350034445479; https://pinterest.slack.com/archives/C05UMECTDDJ/p1751927783667429	
Large ID Embedings + OmniSage + UVE for CLR	Zelun Wang Hongtao Lin	Sitewide repin +0.58% HF num pin click +0.39% HF repins +1.18% HF grid click +0.40% HF long clickthroughs +0.88% HF interest repins +27.77%	N/A	Introduced pin id embeddings	[LR]: Large id embeddings and latest features for Conditional Learned Retrieval	https://pinterest.slack.com/archives/C05UMECTDDJ/p1747950234242069	
Negative Signals to block Recommended Interests	Jenny Liu Lingzhi Luo	-12.79% volume decrease in HF Interest Hides	$4.5k	Improve user experience by filtering out interests that a user has given negative feedback on (extracted from pin hides), with minimal cost.	[LR] homefeed_topics_negative_signal_filter_exp	https://pinterest.slack.com/archives/C05UMECTDDJ/p1748023005202089	
Product Quantization in Learned Retrieval	Hanlin Lu Yuke Yan Bowen Zhou Bella Huang	Non Shopping Click Out Sessions, MECE (SSv2): +0.44% Site-wide Repins: +0.53% Homefeed Clickthroughs: +0.73%	$148k	Product Quantization serves as a foundation for more optimizations in Manas clusters, which is expected to have a 15-20% CPU saving.	[LR] Product Quantization in Homefeed Learned Retrieval	https://pinterest.slack.com/archives/C05UMECTDDJ/p1748377362047569	
Homefeed Timeout Tuning	David Wang J.J Hu	Homefeed Repins +0.47% Homefeed Num Pin Click +0.19% Homefeed Clickthroughs +0.45%	$3.5k cost savings	• Increase the overall system stability in case of a CG incident	hf_timeout_tuning	https://pinterest.slack.com/archives/C05UMECTDDJ/p1750100138453089	
Multi-Embedding Sequence Tuning	Yuke Yan	Save, Revisitation, Social and Download Sessions (SSV2): +0.07% (unique users) Intent Expressions Only Sessions (SSV2): +0.38% HF Closeups: +0.75% HF Repins: +0.46% HF Clickthroughs: +0.52% HF Long Clickthroughs: +0.90% HF Downloads: +0.52%	$12k cost savings	Allows us to use longer sequence (L200 --> L350) and incorporate more user information.	[LR] Homefeed Multi-embedding Retrieval: Sequence Tuning	https://pinterest.slack.com/archives/C05UMECTDDJ/p1750181539495689	
Board CLR to replace Board-based CG budgets	Devin Kreuzer Hongtao Lin Bowen Deng	Engagement/Retention wins: Global MAU: +0.06% (+140k users) Global WAU: +0.08% (+120k users) Total SSv2: +0.18% SRSD SSv2: +0.32% HF Pin Impressions: +0.54% HF Repin Volume: 1.01% HF Repin Propensity: +1.19% HF Close Up Volume: +1.19% HF Downloads: +2.15% HF Screenshots: +2.64% Homefeed Use Case Adoption: +0.78% Homefeed WAU with 2+ use cases Propensity: 0.82% Time Spent: +0.23% Homefeed Timespent: +0.70%	$50k cost savings	First successful application of CLR beyond Interests on HF Grid, establishing the fact that we can continue to expand CLR use cases outside of just Interests to have large metric impact. Establishes a lot of unity and ML code needed to expand to further conditions, and widens the impact of CLR modelling improvements greatly.	[LR] Board CLR on HF Grid	https://pinterest.slack.com/archives/C05UMECTDDJ/p1750447116383969	
Multi-Embedding Non-decayed Pin Perf Deprecation	Zihao Chen Yuke Yan Bella Huang	Total Successful Sessions (SSV2): +0.12% Intent Expression only Sessions, MECE (SSV2): +0.08% Save, Revisitation, Social and Download Sessions (SSV2): +0.11% Non-Shopping Clickout Sessions: +0.38% Shopping Clickout Sessions: +0.21% HF Closeups: +0.28% HF Repins: -0.11% HF Clickthroughs: +0.85% HF Long Clickthroughs: +0.97%	$4k cost savings		[LR] Homefeed Multi-embedding Retrieval: Deprecation of Non-decayed Pin Perf Features on Pin Tower	https://pinterest.slack.com/archives/C05UMECTDDJ/p1751390593806349	

Name	Eng POCs	Metrics Impact (SSv2, HF, Fresh, MAU/DAU), if any	Cost Savings, if any	Foundational Impact (3 sentences max in bullet form), if any	LR Doc	Link to Slack Proposal and Discussion	(WIP) Notes	
New User Onboarding Revamp	Jenny Liu Lingzhi Luo Zihao Chen + many folks on Activations			Significant overhaul of the NUX onboarding process, phasing out topic picker in favor of use cases and sample pins that more accurate showcase what Pinterest is like.	NUX Revamp MVP LR	https://pinterest.slack.com/archives/C073LHUATAB/p1750710687360639		
Pixie CG Increased Fresh Edge Weights	Wei-Ting Lin Bella Huang				[LR] hf_ipfy_fresh_edge	https://pinterest.slack.com/archives/C05UMECTDDJ/p1752096308849439		
NUX Navboost P2P Query Expansion GSS Migration	Jenny Liu	Neutral	$500k cost savings (as part of P2P's efforts)	Migration of HF's use case of query expansion for Navboost P2P (used in NUX) to GSS, contributing to P2P being able to deprecate a Manas cluster.	[LR] hf_navboost_query_expansion_gss_migration	https://pinterest.slack.com/archives/C05UMECTDDJ/p1752187888854679		
Swish Activation and Cosine Learning Rate Schedule in Multi Embedding	Hanlin Lu Bella Huang Yuke Yan	Total Successful Sessions (SSv2): +0.10% Intent Expressions Only Sessions, MECE (SSv2): +0.24% Non Shopping Click Out Sessions, MECE (SSv2): +0.65%, +0.70% (unique users) Shopping Click Out Sessions (SSv2): +0.22% (unique users) Homefeed Num Pin Click: +0.80%, +0.09% (unique users) Homefeed Clickthroughs: +1.28%, +2.05% (unique users) Homefeed Long Clickthroughs: +0.74%, +2.25% (unique users)	NA	Swish activation and cosine learning rate schedule make Multi Embedding model training smoother and converge faster.	[LR] Swish activation and Cosine Learning Rate in Homefeed	https://pinterest.slack.com/archives/C05UMECTDDJ/p1752597482418049		
BMI Migration to GULP backend	Devin Kreuzer	Neutral	Neutral	Establishes the first HF-powered surface on GULP, a top company priority for next year. Opens the opporunity for our team to have increased scope outside HF Grid.	GULP BMI Migration: H1 2025 Update	https://pinterest.slack.com/archives/C05UMECTDDJ/p1752707334520919		
[LR] Homefeed Multi-Embedding: Time Embeddings and C	Yuke Yan	**All Users** Total Successful Sessions (SSV2): +0.13% Intent Expression only Sessions, MECE (SSV2): +0.16% Save, Revisitation, Social and Download Sessions (SSV2): +0.33% Site-wide Actions Pin Impressions: +0.21% Closeup(s): +0.18% Repins: +0.72% Homefeed: HF Pin Impressions: +0.25% HF Closeups: +0.48% HF Repins: +1.44% HF Downloads: +0.86% **Non-Core / Casual Users** Total Successful Sessions (SSV2): +0.31% Intent Expression only Sessions, MECE (SSV2): +0.31% Save, Revisitation, Social and Download Sessions (SSV2): +0.36%		First successful adoption of temporal information in retrieval sequence modeling. First adoption of closeup sequence in retrieval, which have been considered too noisy to use.	[LR] Homefeed Multi-Embedding: Time Embeddings and Clo	https://pinterest.slack.com/archives/C05UMECTDDJ/p1753910233721999		
See More in HF Pin-to-Pin Recs	Jenny Liu	(Low Traffic metrics) +3.47% Homefeed Grid Clicks +1.36% Homefeed Repins (Unique Users) +4.45% Downloads on HF	N/A	First adoption of the new See More action in HF Retrieval, ensuring that from day 1 the new action has an impact on pinners' HFs to build trust with the new action.	[LR] hf_see_more_recs	https://pinterest.slack.com/archives/C05UMECTDDJ/p1754437085563129		
ME Fresh Expanded Neardup Injection	Zihao Chen Yuke Yan	Overall freshness Pins 28d Fresh Impressions: +0.46% Pins 28d Fresh Closeups: +0.83% Pins 28d Fresh Repins: +0.91% Pins 28d Fresh Long Clickthroughs: +0.44%	N/A		[LR] Homefeed Multi-Embedding: Expanded Neardup Freshn	https://pinterest.slack.com/archives/C05UMECTDDJ/p1759769828063359		
ME Fresh Pins Injection	Zihao Chen Yuke Yan	Overall freshness Pins 28d Fresh Impressions: +0.46% Pins 28d Fresh Closeups: +0.83% Pins 28d Fresh Repins: +0.91% Pins 28d Fresh Long Clickthroughs: +0.44%	N/A		[LR] Homefeed Multi-Embedding: Fresh Pins Injection	https://pinterest.slack.com/archives/C05UMECTDDJ/p1755792006754419		
ME PinCLIP Embedding Feature Adoption	Chuxi Wang Zihao Chen Yuke Yan	Overall freshness Pins 28d Fresh Impressions: 0.35% Pins 28d Fresh Repins: 0.73% Pins 28d Fresh Neardup Impressions without Ads: 0.96% Pins 28d Fresh Neardup Repins without Ads: 1.23%	N/A		[LR] Multi-Embedding PinCLIP Embedding Feature Adoption	https://pinterest.slack.com/archives/C05UMECTDDJ/p1755878411983569		
Homefeed L1 Utility	David WangJ.J Hu	Save Sessions, Non-MECE (SSv2): volume +0.12%, propensity +0.15% Repins (users): +0.12% Homefeed - Pin Impressions (users): +0.06% - Num Click: +0.25%, (users) +0.11% - Repins: +0.25% - Pin Hides: -3.46%, (users) -0.97% - Downloads: +0.52% - Screenshots: +0.78%, (users) +0.47%	$115k cost savings	• Add a new utility layer before Pinnability with bussiness control including diversity, shopping, fresh • Redefine L1 presort layer after LWS with throughld diversity control. • Increased parallelism for IPG, pre-score filtering through mini-batch provides substantial latency savings (P99 saving 35ms)	[LR] hf_l1_utility	https://pinterest.slack.com/archives/C05UMECTDDJ/p1756830835739129		
PinnerSpark Interest In Homefeed Conditional Learned Re	Chuxi Wang Jenny Liu Lingzhi Luo	SSv2: Total Successful Sessions (SSV2): 0.08% in Unique Users Intent Expression only Sessions, MECE (SSV2): 0.14% in Actions/User, 0.10% Save, Revisitation, Social and Download Sessions (SSV2): 0.11% in Unique U Homefeed Key Actions: HF Pin Impressions: 0.09% in Unique Users HF Closeups: 0.23% in Actions/User, 0.09% in Unique Users HF Repins: 0.23% in Unique Users HF Long Clickthroughs: 0.61% in Unique Users	NA		[LR] Multi-Embedding PinCLIP Embedding Feature Adoption	https://pinterest.slack.com/archives/C05UMECTDDJ/p1757021202811669		
Text embeddings for Conditional Learned Retrieval	Zelun Wang Chuxi Wang	Total unsuccessful sessions (SSv2) -0.19% Num Explored Pincepts +0.16% Num Explored New Pincepts +0.28%	N/A	Support text emb as a condition in CLR models	[LR]: Text embeddings for Conditional Learned Retrieval	https://pinterest.slack.com/archives/C05UMECTDDJ/p1757446572741299		
Homefeed Aperture Flink Migration	Yuke Yan	Neutral	$82k cost saving	First step towards full migration from PinPin workflows, saving cost and removing dependency on a legacy, hard-to-maintain system.	LR: Hf_Aperture_Realtime	https://pinterest.slack.com/archives/C05UMECTDDJ/p1757536190743189		
Shopping Corpus and CG Size tunning	Wei-Ting Lin J.J Hu	Non-head TML TP Impressions (homefeed): 1.21% TMPL (Excl. Etsy) Impressions (Homefeed, Non-Ad): 7.30% TMPL (Excl. Etsy) 5s+ Outbound Clicks (Homefeed, Non-Ad): 3.87% TMPL (Excl. Etsy) 5s+ Outbound Clicks (Overall, Non-Ad): 1.29% TMPL outbound click rate (Homefeed): 4.22% GPS +0.38%	N/A	• Merge the current "regular" and "TML" corpora (which is approximately the "explored" corpus) • At (manas) indexing time, we apply TML filtering to build a separated TML index for TML retrieval • At serving time, the "regular CG" fetches from the "merged" corpus, and "TML CG" fetches from the "filtered" TML items	[LR] hf_shopping_corpus_cg_size_tuning_exp	https://pinterest.slack.com/archives/C05UMECTDDJ/p1758919208235309		
Migrate Multi-Embedding Training data to Iceberg UID Sort Data	Matthew Lawhon Yuke Yan	Save, Revistation, Social and Download Sessions (SRSD, SSv2): +0.23% Homefeed Num Pin Click: +0.65% Downloads on HF: +2.16% Screenshots on HF: +1.68%	$330k cost saving	Allows ME to train on the same data as Pinnability, unifying the training data pipeline for the retrieval-ranking funnel, saving massive cost saving and improving efficiency.	LR: hf_multiemb_uidsort	https://pinterest.slack.com/archives/C05UMECTDDJ/p1759244355967909		
Incorporate UOEv3d, ProxyUOEv3 and DERM in Homefeed Multi-Emb Learned Retrieval	Hanlin Lu Yuke Yan	Total Successful Sessions (SSv2): +0.08% Intent Expressions Only Sessions, MECE (SSv2): +0.11% Save, Revisitation, Social and Download (SRSD) Sessions, MECE (SSv2): +0.02% Non Shopping Click Out Sessions, MECE (SSv2): +0.50%, +0.20% (unique users) Shopping Click Out Sessions (SSv2): +0.02%	NA	This is the first time that UOE V3 features brought positive impacts on Multi Embedding models, which enabled future developments on offsite signals on L2R models, such as UOE module and soft gating.	[LR] Proxy UOEv3, Offsite PinnerSageV3d and Decoupled E	https://pinterest.slack.com/archives/C05UMECTDDJ/p1759273867020149		
Single CLR Scorpion Request	Devin Kreuzer Piyush Maheshwari	N/A	$350k Cost Savings, reducing cost if infe	First time we combine all condition model requests into one, permits future foundational changes for GPU serving and Request-level feature optimizations (attention masking etc.)	[LR] CLR Serving Optimization: Single Model Request	https://pinterest.slack.com/archives/C05UMECTDDJ/p1759772029867969		
Multi-Embedding: Serve Viewer Tower on GPU	Matthew Lawhon Yuke Yan	Homefeed Clickthroughs: +0.92% Homefeed Long Clickthroughs: +1.37% Model Latency (P99): 99ms --> 17ms Model Latency (P50): 60ms --> 13ms	$150k cost saving after full migration anc	First time to serve a retrieval model on GPU. Reduce latency by 72%, and unlocking capacity for much more complex, larger models.	LR: homefeed_multi_emb_gpu_serving	https://pinterest.slack.com/archives/C05UMECTDDJ/p1759848557880399		
CLR Fresh Pins Injection	Zili Li Devin KreuzerZihao Chen Yuke Yan	Overall freshness Pins 28d Fresh Impressions: +1.01% Pins 28d Fresh Closeups: +1.45% Pins 28d Fresh Repins: +0.99% Pins 28d Fresh Repins + Long Clickthroughs: +0.87% Pins 28d Fresh Impressions without Ads: +1.64% Pins 28d Fresh Closeups without Ads: +1.56%	N/A	N/A	[LR] Homefeed Conditional Learned Retrieval: Fresh Pins Inj	https://pinterest.slack.com/archives/C05UMECTDDJ/p1759860794391089		
HF See More Recs w/ Time Weights	Jenny Liu	(Low Traffic Metrics) +0.48% Save, Revisitation, Social and Download (SRSD) Sessions, MECE (S +0.54% Homefeed Pin Impressions	N/A	Improvement on the original See More recs launch for future-proofing as actions increase in volume, ensuring both immediate impact on the feed (using most recent action) and long-term impact (sampling actions across the user's See More events)	[LR] hf_see_more_recs_time_weights	https://pinterest.slack.com/archives/C05UMECTDDJ/p1760053649721209		
PinCLIP Image Embedding Feature Adoption in CLR	Zelun Wang	BMI landing page repins: +2.23% BMI landing page screenshots: +2.11% Sitewide Unique User Fresh repins: +0.27% HF Unique User Fresh repins: +0.30% HF Interest Repins (unique): +0.37% Unsuccessful SSv2: -0.13% HF Long clickthroughs: +0.82% Explored New Pincepts: +0.09%	N/A	N/A	[LR] PinCLIP Image Embedding Feature Adoption in CLR	https://pinterest.slack.com/archives/C05UMECTDDJ/p1760465107984989		
Demegraphic features in PBTY	Zihao Chen	0.9% 14d WAU	N/A	This feature addition can be applied to all HF CGs to help surface-level LSU engagement & retention	[LR] Adding Viewer Demo Features in Pinnability	https://pinterest.slack.com/archives/C05UMECTDDJ/p1761078628991149		
NUX TT for All NUX & Resurrected users	Zihao Chen	Session: Non Shopping Click Out Sessions (SSv2): 0.62% Homefeed engagement: HF Repins: +0.65% Unique Users HF Closeups: +0.88% Actions/User, +0.32% Unique Users HF Clickthroughs: +2.37% Actions/User, +1.64% Unique Users Sitewide engagement:			[LR] NUX-TT Expansion: Coverage for All NUX & Resurrecte	https://pinterest.slack.com/archives/C05UMECTDDJ/p1765217743107349		
NUX TT for NUX Revamp	Zihao Chen	0.8% 14d WAU HF Closeups: 2.54% Actions/User, 0.47% Unique Use HF Repins: 0.92% in Actions/User, 1.37% Unique Users HF Clickthroughs: 4.98% in Actions/User, 2.25% Unique Users HF Long Clickthroughs: 10.29% in Actions/User, 4.57% Unique Users HF Downloads: 2.73% in Actions/User, 1.46% Unique Users	N/A	A new patform dedicated to drive NUX/ LSU engagement & retention.	[LR] NUX Revamp: Two-Tower Candidate Generator (NUX-T	https://pinterest.slack.com/archives/C05UMECTDDJ/p1761685357062779		
CLR Expanded Neardup Freshness Injection	Zili Li Zihao Chen Yuke Yan	Neardup freshness Pins 28d Fresh Neardup Impressions without Ads: +1.08% Pins 28d Fresh Neardup Closeups without Ads: +1.04% Pins 28d Fresh Neardup Repins without Ads: +1.11% Pins 28d Fresh Neardup Repins + Long Clicks without Ads: +1.08%	N/A	N/A	[LR] Homefeed Conditional Learned Retrieval: Expanded Ne	https://pinterest.slack.com/archives/C05UMECTDDJ/p1761921547104649		
LWS Calibration Dataset Iceberg Table Migration	Yidi WangHedi Xia	N/A	~$220k annual S3 cost savings	Improve L1/L2 Alignment Provided an example to adopt the iceberg datasets in LWS	[LR] LWS Calibration Dataset Iceberg Table Migration	https://pinterest.slack.com/archives/C05UMECTDDJ/p1762301780573529		
Learnable Context-Sensitive Gating for Adopting UOE anc	Yidi WangHedi Xia	SSv2: Total Successful Sessions (SSv2): +0.13% Intent Expressions Only Sessions, MECE (SSv2): +0.13% Save, Revisitation, Social and Download (SRSD) Sessions, MECE (SSv2): +0 Homefeed: Homefeed Pin Impressions: +0.22% Homefeed Num Pin Click: +0.21%. Timespent overall: Total time spent in App: +0.21% Homefeeds: +0.23%	N/A	provided an idea to safely adopt the implicit coverage features without metric tradeoffs	[LR] Learnable Context-Sensitive Gating for Adopting U	https://pinterest.slack.com/archives/C05UMECTDDJ/p1760731988281299		
BMI on GULP: Migrated the CG (including CLR) and Blen	Devin Kreuzer	N/A	N/A	Establishes the first HF-powered surface on GULP, a top company priority for next year. Opens the opporunity for our team to have increased scope outside HF Grid.	[LR] Board More Ideas on GULP	https://pinterest.slack.com/archives/C05UMECTDDJ/p1762464743739679		
Interest Service Deprecation	Alok Malik J.J Hu	N/A	$424k cost saving	Completely deprecate interest service which supports topicfeed, SEO, notif, bestpin2/3, and migrate the new servig into unity-homefeed	Interests Service Deprecation	https://pinterest.slack.com/archives/C0145F4SW9G/p1763678558103699		
Embedding quantization to HF ItemSage CG	Yujiao Guo J.J Hu	N/A	$29.6k cost saving	unblock further quantization work to optimize manas shopping cluster host optimization workstream	LR: itemsage_quantization	https://pinterest.slack.com/archives/C05UMECTDDJ/p1754606731956039		
HF Shopping Load Tuning using Reinforcement Learning	Alok Malik Mehdi Ben Ayed Charlie Tian (He Him)	Sitewide Repins: +.51% Sessions >5m: +.32% Time Spent: +.26% Pin Impressions: +.23% Ad Impressions: +.22% Homefeed Closeups: +.38% Homefeed Long Clicks: +.98% Homefeed Use Case Adoption: +.47% Time Spent on Homefeed: +.28% TWP impression on Homefeed: -7.95% Golden Path Users: Neutral (+.13% non stat sig) Infra Cost Savings $258k (i.e. approx -20% total cost reduction - a finer graine	$260k cost savings	Introduces a new layer in recommender system to make use of online serving infra more efficient	[LR] HF Shopping Load Tuning using Reinforcement Learnin	https://pinterest.slack.com/archives/C05UMECTDDJ/p1763511551356659		
Conditioned User Sequence in CLR	Devin Kreuzer Piyush Maheshwari	Total Successful Sessions (SSv2): +0.08% US Market: +0.19% SRSD (SSv2): +0.10% US Market: +0.26% Shopping Click Out Sessions (SSv2): +0.29% Homefeed: Homefeed Repins: +0.65% (Propensity: +0.44%) Homefeed Pin Screenshot Propensity: +0.54% Homefeed Pin Reactions: +0.92% Homefeed Pin Shares: +1.72% Total Time Spent in App: +0.12% Time Spent on Site: +0.33% Users > 30 mins: +0.29% Homefeed timespent: +0.27% Board More Ideas (strongly diluted): Repin Propensity: +0.49%	Neutral	Introduces user sequence modelling to CLR for the first time, establish a baseline for the team to work on. This is a foundational improvement which will steer most of the modelling work on CLR for next year. All the FM code will be built on top of this change.	[LR] Unified CLR: Conditioned User Sequence Module	https://pinterest.slack.com/archives/C05UMECTDDJ/p1763420412291149		
Pin-based Conditional Learned Retrieval	Zili Li Devin Kreuzer J.J Hu	SSv2 Total Successful Sessions (SSV2): +0.13% Intent Expression only Sessions, MECE (SSV2): +0.02% Save, Revisitation, Social and Download Sessions (SSV2): +0.16% Non-shopping Clickout Sessions Sessions (SSV2): +0.45% Shopping Clickout Sessions (SSV2): +0.29% Homefeed Homefeed Pin Impressions: +0.18% Homefeed Num Pin Click: +0.64% Homefeed Repins: +0.88% Homefeed Clickthroughs: +1.32% Homefeed Long Clickthroughs: +2.41%	50k cost increase	Expands Pins as a Condition to the CLR model, opening the door for further pin-based CLR retrieval with UIC signals, and on other surfaces inluding Notif-HF landing page effort. Large metric gains fueling further investment in CLR	[LR] Pin CLR on HF Grid	https://pinterest.slack.com/archives/C05UMECTDDJ/p1765288812801759		
Cost Optimization for HF Request after Chunk Invalidatior	Raymond Hsu J.J Hu Dafang He	N/A	$111K-$140K cost savings		[LR] hf_subchunk_cg_ablation	https://pinterest.slack.com/archives/C05UMECTDDJ/p1765413405679339		
HF Pin Selection V1 UCPD Data Deprecation	Jenny Liu J.J Hu	N/A	$183k cost savings		[LR] hf_ucpd_ps_index_lr + hf_ucpd_ps_index_clr_24	https://pinterest.slack.com/archives/C05UMECTDDJ/p1765828625983819		
User Interest Cluster (UIC) in CLR	Chuxi Wang Yuke Yan	SSV2 Total Successful Sessions (SSV2): 0.07%, +0.11% (in unique users) Save, Revisitation, Social and Download Sessions (SSV2): +0.19%, +0.13% (in unique users) Non-Shopping Clickout Sessions (SSv2): +0.41% Site-wide Actions Pin Impressions: +0.02%, +0.10% (in unique users) Repins: +0.17% Homefeed: HF impressions: +0.11%, +0.09% (in unique users) HF Repins: +0.69% HF Clickthroughs: +0.79% HF Long Clickthroughs: +1.28% HF Downloads: +2.87% HF Screenshots: +2.02% Pin Share Completed on HF: +1.34% Retention DAU: +0.17% WAU: +0.14%	$332k cost saving	We successfully built the UIC (user interest cluster) V2 signal, and successfully integrate it with the CLR model. This provides a massive benefit in SSV2, HF engagement, retention, and diversity. An important milestone for retentive recommendation workstream. This lays the foundation for the future exploration work, as we can feed the predicted UIC through the same pipeline.	[LR] Retentive Recommendations: UIC signals in CLR			

Name	Eng POCs	Metrics Impact (SSv2, HF, Fresh, MAU/DAU), if any	Cost Savings, if any	Foundational Impact (3 sentences max in bullet form), if any	LR Doc	Link to Slack Proposal and Discussion	(WIP) Notes	
UPP CLR Feature Alignment: Annotations v7 Upgrade + L	Sophia ZhuDevin Kreuzer	Freshness Fresh Impressions: +0.80% Fresh Repins: +2.50% Pins 28d Fresh Neardup Impressions without Ads: +1.61% Pins 28d Fresh Neardup Repins without Ads: +2.65% Pins 28d Fresh Neardup Long Clicks without Ads: +1.12% Homefeed Homefeed Closeup Propensity: +0.10% Homefeed Repin Propensity: +0.20% Homefeed Downloads: +0.68% Homefeed Screenshot Propensity: +0.54% Homefeed Interest Pin Impressions: +4.06% Homefeed Interest Repins: +6.92% Homefeed Interest Grid Clicks (Closeups): +5.22% Homefeed Interest Clickthroughs: +5.53% Homefeed Interest RCs: +6.78% Homefeed Use Case + Timespent Homefeed Timespent: +0.19% Homefeed Use Case Adoption: +0.44% Impression Diversity: +0.33% Homefeed WAU with 2+ use cases: +0.20%		In this work for the UPP (Unified Personalization Platform) project, we apply two changes (that are migrating pinnability_annotation_v5 to v7 and deprecating three pin features) to the training datasets for homefeed CLR model, as an effort of feature alignment for a unified base CLR model across both homefeed and notif surfaces. This will help align base CLR model training in the UPP project and reduce the extra work of logging redundant features, simplifying unification of CLR model across homefeed and notif surfaces.	[LR] UPP CLR Feature Alignment: Annotations v7 Upgrade +	https://pinterest.slack.com/archives/C05UMECTDDJ/p1765218319127799		
Viewer Demographic Features Adoption in CLR	Sophia Zhu Devin Kreuzer	Key Metrics for Casual Users: Non Shopping Click Out Sessions, MECE (SSv2): +0.46% Total Unsuccessful Sessions (SSv2): -0.27% 5s+ Outbound Clicks: +0.43% Homefeed Pin Impressions: +0.27% Homefeed Num Pin Click: +0.50% Homefeed WAU with 2+ use cases: +0.26% Board Creates: +0.85% More Ideas Ad Closeups: +0.72% Board Compact View (Mobile Only): +18.93% Key Metrics for All Users: Board Creates: +0.30% Total Unsuccessful Sessions (SSv2) Propensity: -0.15% Homefeed WAU with 2+ use cases (Propensity): +0.19% More Idea Num Outbound Clicks: +0.75%		In this work, we adopted two viewer demographic features (i.e. pinnability.viewer.onlineAge and pinnability.viewer.onlineGender) for the CLR training datasets. By improving the personalization, we are able to see metric gains for all users and especially for low-signal user groups, such as casual users.	[LR] Viewer Demographic Features Adoption in CLR	https://pinterest.slack.com/archives/C05UMECTDDJ/p1765401985776619		
Onboard BoardCLR to LWS and L1 Utility	Yali Bian Hedi Xia J.J Hu	NA	~$21K/year savings		[LR] Onboard BoardCLR to LWS and L1 Utility	https://pinterest.slack.com/archives/C05UMECTDDJ/p1767991644190129		

PLEASE UPDATE THE 2026 TAB INSTEAD THANK YOU!!

Metric	Cumulative Sum
Increase in SSv2 (Total Successful Sessions or otherwise	2.12%
Increase in Save, Revisitation, Social, and Download (SR!	2.27%
Homefeed Repins	22.52%
Weekly Active Users (WAU)	0.33%
Daily Active Users (DAU)	0.17%